29/10

MATHS

1.

Cost of living	No. of we.	cf
1400 - 1550	8	8
1550 - 1700	15	23
1700 - 1850	21	44
1850 - 200	8	52

$n = 52$

$\frac{n}{2} = 26$

1700-1850 is ~~middl~~ median class.

2. $2\pi r = 22$

$2 \times \frac{22}{7} \times r = 22$

$r = \frac{7}{2}$ cm

$\theta = 90^\circ$

area of quadrant $= \frac{\pi r^2 \theta}{360}$

$= \frac{\cancel{22}^{11}}{\cancel{7}} \times \frac{\frac{\cancel{7}}{\cancel{2}} \times \frac{7}{2} \times \cancel{90}}{\cancel{360}\ 4}$

$= \frac{77}{8}$ cm^2.

$\therefore$ area of quadrant is $\frac{77}{8}$ cm^2.

3 Modal size of shoes is 5.

4. $R = 5$

$r = 4$

$\theta = 30$

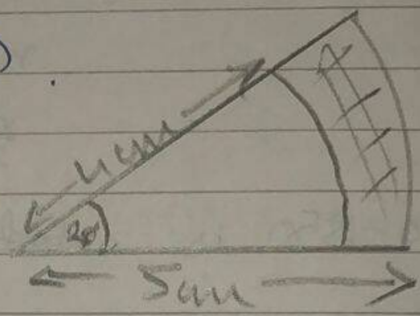
Area of shaded region.

$$\frac{\pi\theta}{360}(R^2 - r^2).$$

$$\frac{\frac{22}{7} \times \cancel{30}}{\cancel{360}_{12}} \times (25 - 16).$$

$$\frac{\cancel{22}^{11}}{7 \times \cancel{12}} \times \cancel{9}^{3}$$

$$\frac{33}{14} \text{ cm}^2.$$



$\therefore$ area of shaded region is $\frac{33}{14}$ cm^2

5 Mode = 35.3

~~Median~~ Mean = 30.5

3 Median − 2 Mean = Mode

3 Median = Mode + 2 Mean

$$\text{Median} = \frac{35.3 + (30.5)2}{3}$$

$$\text{Median} = \left(\frac{35.3 + 61}{3}\right)$$

$$\text{Median} = \left(\frac{96.3}{3}\right)$$

Median = 32.1

$\therefore$ Median is 32.1.

$\theta = \left[\frac{360}{12}\right] = 30^\circ$

$r = 14$

$\text{Area} = \frac{\pi r^2}{360} \times \theta$

$= \frac{22}{7} \times 14 \times 14 \times \frac{30}{360}$

$= \frac{154}{3} \text{ cm}^2$

Area swept by minute hand is $\frac{154}{3}$ cm^2.

Pocket Money	No of students.
0 - 20	2
20 - 40	2
40 - 60	3
60 - 80	12 (f_0)
80 - 100	18 (f_1)
100 - 120	5 (f_2)
120 - 140.	2

h (Class size) = 20.

$l = 80$.

$\text{Mode} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] h$.

$= 80 + \left[\frac{18 - 12}{36 - 12 - 5}\right] 20$

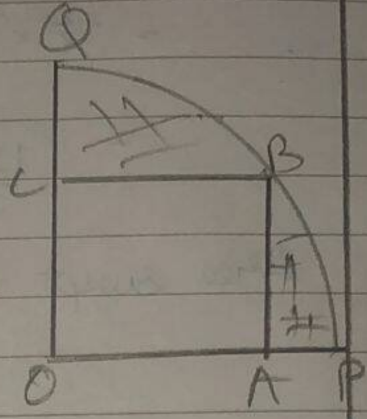
$= 80 + \left[\frac{6}{19}\right] 20$

$= 80 + 13.3$

$= 93.3$ $\therefore$ Modal Money is 93.3

5. Join BO.
so BO is radius 'r'
$\angle C = 90°$ (since line from center is $\perp$ to chord)

Applying pythagoreas theorem.



$BC^2 + CO^2 = BO^2$

$BC^2 + BC^2 = BO^2$

$\sqrt{2OA^2} = BO$

$\sqrt{2 \times 400} = BO$

$20\sqrt{2}\ cm = BO$

$20 \times 1.414 = BO$

$20\sqrt{2} = 28.28 = BO = r.$

area of OABC $= (20)^2$
$= 400\ cm^2$

area of quadrant $= \frac{\pi r^2}{360}\theta = 3.14 \times \frac{20 \times \sqrt{2} \times 20\sqrt{2}}{\cancel{360}\ 4} \times \cancel{90}$

$= 3.14\ \frac{\overset{100}{\cancel{400}} \times 2}{4}$

$= 628\ cm^2.$

area of shaded region $= 628 - 400$
$= 228\ cm^2$

$\therefore$ area of shaded region is $228\ cm^2$

9.

class	frequency	$x_i = \frac{UL+LL}{2}$	$d_i = x_i - a$	$u_i = \frac{d_i}{h}$	f_iu_i
0-20	17	$\frac{0+20}{2} = 10$	-40	-2	-34
20-40	p	30	-20	-1	-p
40-60	32	50 (a)	0	0	0
60-80	24	70	20	1	24
80-100	19	90	40	2	38
	$f_i = 93+p$				$\Sigma f_iu_i = 28-p$

$h = 20.$

$$\text{Mean} = a + \left[\frac{\Sigma f_iu_i}{\Sigma f_i}\right]h$$

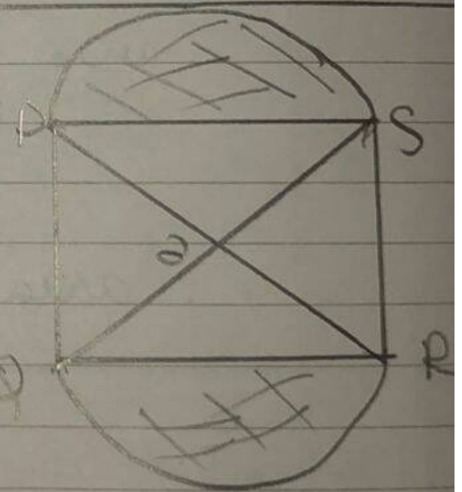
$$\cancel{50} = \cancel{50} + \left[\frac{28-p}{93+p}\right]20$$

$$0 = [28-p]\,20$$

$$p = 28.$$

$\therefore$ p is 28.

10. $PQ = PS \times RS = QR = 42\,cm.$



In ΔSPQ

Applying pythagoreas theorem

$$PS^2 + PQ^2 = QS^2$$

$$\sqrt{(42)^2 + (42)^2} = QS$$

$$42\sqrt{2}\,cm = QS$$

$SO = QO$ ($\because$ diagonal bisect each other at 90°)

$\theta = \angle SOP = 90°$

$$\frac{42\sqrt{2}}{2} = SO = \text{radius} \;--\; r.$$

$$r = 21\sqrt{2}\,cm.$$

2 sector
area of ~~(shaded region)~~ = $2\left[\frac{\pi r^2}{360}\cdot\theta\right]$

$\left[\cancel{= 2\left[\frac{22}{7}\times \right]}\right]$

$\Rightarrow 2\left[\frac{\frac{22}{7}\times 21\sqrt{2}\times 21\sqrt{2}\times \cancel{90}}{\cancel{360}\ 4\ 2}\right]$

$\Rightarrow \frac{22}{7}\left[\frac{21\times 21\times \cancel{2}}{\cancel{2}}\right]$

$= [22][441]/7$

$\frac{9702}{7}$ m² = ~~3~~ $\boxed{1386\,m^2}$

~~(∴ area of shaded region is 9702m²)~~

area of 2.~~(segment)~~ Δ = $2\left[\frac{1}{2}\times PO\times SO\right]$

$= (21\sqrt{2})(21\sqrt{2})$

$= \boxed{882\,m^2}$

area of shaded region =

$1386 - 882$

$\boxed{504\,m^2}$.

∴ area of shaded region is 504m²